Problem Statement & Mission Objectives

Anticipating River Discharge Surges & Mitigating Flash Flood Catastrophes

Team VibeForge

The Hydrological Crisis

HIGH IMPACT

- **Core Challenge:**
  Extreme riverine flood events cause $8B+ in annual US damages and severe life safety risks.
- **Hydrological Complexity:**
  Non-linear catchment response: antecedent rainfall pulses, soil saturation delay, and basin slope create deadly crest surges.
- **Machine Learning Mission:**
  Predict multi-station daily river discharge (discharge_cfs) across 50 diverse US basins with zero temporal leakage.
- **Power-Law Extremes:**
  Standard regression under-predicts rare 90th-percentile flood crests by optimizing for quiescent baseflow.

Project Objectives & Scope

OBJECTIVES

- **Multi-Station Daily Forecasting:**
  Predict discharge_cfs across 36,500 test station-day intervals (2022–2023).
- **Extreme Event Accuracy:**
  Prioritize peak crest predictability (top 10% flow days) over routine low-flow baseflow.
- **Physically Constrained Predictions:**
  Strict enforcement of non-negative discharge rates (discharge >= 0.0 cfs).
- **Real-Time Decision Intelligence:**
  Bridge ML inference to an interactive command center for watershed emergency managers.

Data Architecture & Feature Engineering Pipeline

Harmonizing USGS NWIS Streamflow, Open-Meteo Weather & StreamStats Geomorphology

Team VibeForge

Data Sources & Ingestion

50 STATIONS

- **USGS NWIS Telemetry:** 12 years (2010–2021) of daily discharge observations across 50 diverse US river stations (219,150 records).
- **Open-Meteo Historical Archive:** Daily liquid precipitation (mm), maximum/minimum 2m air temperatures (°C), and volumetric root-zone soil moisture.
- **USGS StreamStats Database:** Static catchment physiography: drainage area (sq mi), mean basin elevation (ft), slope (% grade), and forest cover (%).
- **Strict Chronological Holdout:** Train: 2010-01-01 to 2021-12-31 | Test: 2022-01-01 to 2023-12-31. Absolute zero future leakage.

Engineered Hydrological Signals

19 FEATURES

- **Rolling Cumulative Hyetographs:** 1d, 3d, 5d, 7d, and 14d rolling sums capturing multi-day storm accumulation and delayed runoff.
- **Antecedent Hydrological Lags:** 1d, 2d, and 3d lagged soil moisture & temperature capturing basin saturation threshold and snowmelt dynamics.
- **Runoff Volume Interaction Proxy:** Rainfall-runoff proxy computed as precipitation_mm × drainage_area_sqmi representing total water influx.
- **Station-Isolated Transforms:** All rolling windows & lags computed per-station to eliminate inter-basin data leakage.

Model Architecture & Extreme Event Optimization

Symmetric Oblivious Trees, Native Categorical Hashing & Extreme Sample Weighting

Team VibeForge

Machine Learning Architecture

CATBOOST

- **CatBoost Regressor:**
  Primary architecture: Gradient-boosted decision trees with symmetric oblivious tree structure to prevent overfitting.
- **Native Categorical Encoding:**
  Station ID encoded natively through ordered target statistics without one-hot dimensionality explosion.
- **Hyperparameter Tuning:**
  iterations=600-1500, learning_rate=0.03-0.06, depth=6, loss_function='RMSE'.
- **Hydrologic Bounds Enforcement:**
  Predictions post-processed with np.clip(predictions, 0.0, None) ensuring zero negative flow violations.

Extreme Event Weighting Scheme

2.5X PEAK WEIGHT

- **The Tail Bias Problem:**
  Power-law distribution of flow means 90% of samples are low baseflow; standard loss treats all errors equally.
- **2.5x Peak Loss Weighting:**
  Observations in the top 15% historical discharge receive a 2.5× sample weight multiplier during gradient calculation.
- **Evaluation Metric Suite:**
  40% Overall RMSE + 30% Extreme Event RMSE (Top 10% Flows) + 20% MAE + 10% R^2 Score.
- **Dual Ensembling Strategy:**
  Optional LightGBM blend provides diverse tree splits across high-elevation snowpack catchments.

Results, Validation & Submission Compliance

Comprehensive Performance Benchmarks & 100% Automated Assertion Verification

Team VibeForge

Model Accuracy Benchmarks

VALIDATED

- **Overall Discharge RMSE (40% Weight):**
  2,065.14 cfs — Strong baseline tracking across low and moderate seasonal flow stages.
- **Extreme Event RMSE (30% Weight):**
  3,681.89 cfs — Top 10% peak flow tracking powered by 2.5x sample-weighted gradient penalties.
- **General Prediction Reliability (MAE):**
  782.26 cfs (20% Weight) — Consistent day-to-day volumetric accuracy.
- **Pattern Capture Quality (R^2 Score):**
  0.9980 (10% Weight) — Captures hydrological seasonality and storm hydrograph rises.

Official Submission Compliance

7/7 PASS

- **Exact Row Count (36,500 rows):**
  Matches test.csv row-for-row across all 50 stations and all 730 test dates [PASS].
- **Non-Negative Flow Constraint:**
  Zero negative discharge predictions (Min: 0.00 cfs, Max: 870,435.35 cfs) [PASS].
- **Zero Nulls / NaNs / Infs:**
  100% complete float data with zero missing entries or unhandled infs [PASS].
- **Format & Identifier Preservation:**
  All 47 stations with leading zeros (e.g. '01646500') preserved as strings [PASS].
- **Date Alignment (YYYY-MM-DD):**
  Continuous chronological progression from 2022-01-01 to 2023-12-31 [PASS].

Known Limitations, Ethics & Future Horizons

Transparent Boundary Conditions and Roadmap Toward Real-Time Global Telemetry

Team VibeForge

⚠Known Limitations

TRANSPARENCY

- **USGS Geodatabase Dependency:**
  StreamStats static attributes require US-standardized catchment boundaries; expansion requires local DEM modeling.
- **100-Year Tail Outliers:**
  Tree-based regressors cannot extrapolate beyond historical boundaries during unprecedented atmospheric events.
- **Anthropogenic River Regulation:**
  Dam retention, reservoir spillway releases, and irrigation locks are unmodeled external drivers.
- **Sub-Daily Urban Flash Surges:**
  Forecasts operate on daily timestep; micro-tributary urban flash floods require hourly radar input.

Future Roadmap & Impact

ROADMAP

- **Spatial Graph Neural Networks (GNNs):**
  Model upstream-to-downstream topological river network connectivity with message-passing.
- **Satellite Snowpack Ingestion:**
  Integrate NASA MODIS/Sentinel-2 snow water equivalent (SWE) for improved spring runoff timing.
- **Ensemble Probabilistic Bands:**
  Output quantile loss prediction intervals (10%, 50%, 90%) for actionable uncertainty bounds.
- **Command Center Deployment:**
  Active Next.js 15 dashboard is live and operational with interactive Leaflet mapping & hydrographs.